

The Pool of Beta-lactamase Present in Extended Drug Resistant *Klebsiella pneumoniae* Isolated from Clinical Settings in Dhaka, Bangladesh

Akter S¹, Tabassum T², Khaleque M³, Islam MF⁴, Akhter H⁵, *Begum A⁶

Recently bacterial infections caused by antibiotic resistant bacteria are becoming a threat to the health care system of many countries. Extensively-drug-resistant (XDR) *Klebsiella pneumoniae* are resistant to almost all classes of antibiotics and only a few are for clinical use. Recent emergence, worldwide dissemination and clinical impact of carbapenemase positive isolates persuade us to characterize the clinical *K. pneumoniae* isolates. We tested 647 *K. pneumoniae* that was isolated from different specimens of hospitalized patients in Dhaka, Bangladesh. Various phenotypic characterization, plasmid profiling and PCR assays followed by sequencing were performed. Round the year, 4.33% (n=28) *K. pneumoniae* isolates were found as XDR in addition to carbapenem resistant. Among the XDR isolates, 82.0% (n=23) showed Metallo-beta-lactamase (MBL) phenotype. Only two isolates showed MBL with KPC (*K. pneumoniae* carbapenemase) phenotype. By double disc synergy test, 12 isolates (43.0%) showed extended-spectrum beta-lactamase (ESBL) phenotype in addition. Among the isolates, 60.0% (n=17) were blaTEM positive, 82.0% (n=23) were blaSHV positive and 46.0% (n=13) were blaOXA positive determined by multiplex PCR. blaCTX-M were present in 43.0% (n=12) isolates, which were of type-15 as revealed by sequencing. More than 60.0% (n=17) isolates were positive for blaNDM-1. Multiple genes coexisted in nearly all isolates. None of those were found to carry blaVIM or blaKPC. Twenty-six isolates were found to be strong biofilm former. Among the isolates, 16 harbored multiple plasmids ranging from 19 to 91 KDa; however, correlation between plasmid profile and antibiotic resistance pattern was not evident. Prevalence of carbapenem resistant and ESBL isolates have been previously reported in Bangladesh. Therefore, this study will expand our knowledge about XDR disseminated in Bangladesh and could assist in future therapeutic management.

[Mymensingh Med J 2025 Apr; 34 (2): 428-437]

Key words: Beta-lactamase, drug resistance, *Klebsiella pneumoniae*, Clinical setting

Introduction

The emergence and spread of carbapenem-resistant Gram-negative bacteria are a worldwide emerging public health threat and concern responsible for large number of infections. Treatment options for patients infected with Carbapenem-resistant *Klebsiella pneumoniae* (CRKP) are limited, for they are usually resistant to all beta-lactams and most other classes of antimicrobial agents. Emergence and global dissemination of extended-spectrum beta-lactamase (ESBL) phenotype among bacteria has made carbapenems the last resort antibiotics for clinical use¹. The most common mechanisms of resistance are the production of carbapenem-hydrolysing enzymes, carbapenemases that hydrolyze most β -lactams group of antibiotics. Clinical impact and recent prevalence of infections by carbapenemase-producing bacteria caught the global health attention². CRKPs are frequently associated with severe nosocomial infections and outbreaks³. Before NDM-1 (New Delhi Metallo-beta-lactamase) was reported in

2009, the IMP (active on imipenem) and VIM (Verona integron - encoded Metallo - beta - lactamases) types were the common Metallo - beta - lactamases identified in Enterobacteriaceae⁴.

1. Selina Akter, Department of Microbiology, Jashore, University of Science and Technology, Jashore, Bangladesh
2. Tania Tabassum, Department of Microbiology, University of Dhaka, Dhaka, Bangladesh
3. Marium Khaleque, Department of Microbiology, University of Dhaka, Dhaka, Bangladesh
4. Humaira Akhter, Department of Microbiology, University of Dhaka, Dhaka-1000, Bangladesh
5. Dr Mohammad Faridul Islam, Department of Microbiology, Square Hospitals Limited, Dhaka, Bangladesh
6. *Dr Anowara Begum, Department of Microbiology, University of Dhaka, Dhaka, Bangladesh; E-mail: anowara@du.ac.bd

*for correspondence

Detection of carbapenemase production is difficult due to some carbapenemase-producing isolates demonstrate false susceptible phenotype with high MIC ranges⁵. Clinical and Laboratory Standards Institute (CLSI) has published guidelines for the detection of isolates producing carbapenemases (CLSI document M100)⁶. Newly evolving genes escalates the list of enzymes contributing resistance and limits the accurate detection. The recent development of large-scale *bla* detection method may overcome the limitations⁷ to some extent. Although NDM-1 was first recorded in 2008 in a *K. pneumoniae* strain isolated from a Swedish patient who had traveled to New Delhi⁸ that suggested the isolates producing the NDM-1 enzyme were disseminated in Indian healthcare facilities as early as 2006⁹. In Bangladesh, presence of the NDM-1 gene in *K. pneumoniae* was reported as a predominant isolate from different surveillance studies between 2003 and 2010¹⁰. An early initiative has also been taken to track the possible sources of spread of these organisms in Dhaka¹¹. Lack of implementing standardized protocols for detection of carbapenemase-producing isolates the actual prevalence of these producers is still unknown. It is said that we are on the edge of a global epidemic with carbapenemase-producing isolates, which in the hospital environment is likely to be caused mostly by all types of nosocomial carbapenemase-producing Gram negative Enterobacteriaceae (e.g. KPC, IMP, VIM, NDM and OXA-48)⁴.

Coexistence of beta-lactamases are often found widely distributed, especially the combinations of OXA-48, SHV, TEM and CTX-M-type. A high prevalence of the NDM-1 in CRKP has also co-existed with the above¹. Biofilm-forming ability of *K. pneumoniae* strains makes the bacteria up to 1,000 times more resistant to antibiotic therapy¹². The present study targeted the phenotypic and molecular characterization of extensively drug-resistant (XDR) *K. pneumoniae* isolates found in clinical setting in Bangladesh. XDR is the term defined by standardized international terminology created in joint initiative by the European Centre for Disease Prevention and Control (ECDC) and the Centers for Disease Control and Prevention (CDC)¹ for enterobacteriaceae isolates which remain susceptible to only one or two categories of antimicrobials respectively.

Methods

Six hundred and forty seven *K. Pneumoniae* strains were isolated over a year from January 2019 to January 2020 from different clinical specimens obtained from patients who attended a tertiary care private hospital in Dhaka, Bangladesh. Samples mostly comprised of urine, blood, sputum, pus, suction tip, tracheal aspirates, broncho-alveolar lavage and CV line tip.

Processing of samples

Blood samples were incubated at 37°C overnight in Blood Culture broth (BD BACTEC™ with BACTEC 9240 System). Culture broth was inoculated on MacConkey agar, as well as chocolate agar plates (in 5.0% CO₂ supply) and incubated at 37°C for overnight. The *K. pneumoniae* isolates were picked as a pure and predominant colony growing on media. In the hospital, the organisms were identified based on colony morphology, examination under light microscope followed by Gram staining and some specific biochemical reactions. Selected isolates were transported to the Laboratory in the Department of Microbiology, the University of Dhaka within an hour in a cool box and performed further works. Ethical approval was deemed unnecessary because the laboratory received isolated cultures exclusively, without direct involvement with patient specimens or patient care and thus Isolates were verified by Gram reaction, biochemical reaction pattern and finally biochemical profiling for several randomly selected organisms using API 20E kit (Biomérieux, France)¹³.

Antibiotic susceptibility testing

For the antibiotic sensitivity profiling, commercially available antibiotic discs of Amikacin (30µg), Amoxicillin (10µg), Ampicillin (10µg), Cefepime (30µg), Cefixime (5µg), Ceftriaxone (30µg), Cefuroxime (30µg), Chloramphenicol (30µg), Ciprofloxacin (5µg), Colistin (5µg), Gentamycin (15µg), Imipenem (10µg), Meropenem (10µg), Nitrofurantoin (300µg), Tetracycline (30µg), Tobramycin (10µg) and Trimethoprim-Sulphamethoxazole (25µg) were used. The National Committee for Clinical Laboratory Standards (NCCLS) guideline^{6,14} was followed and the test was performed according to modified Kirby-Bauer's disc diffusion method¹⁵ on Mueller-Hinton Agar (Hi-Media, India) media. *Escherichia coli* ATCC 25922 was used as control throughout the study.

Phenotypic detection of carbapenemase producing isolates

Among the *K. pneumoniae* isolates, 28 were phenotypically resistant to Carbapenem in Kirby Bauer agar-disc-diffusion method¹⁴ and were selected for modified Hodge test¹⁶, a phenotypic screening method for carbapenemase activity. The carbapenemase production was confirmed when the test isolate allowed growth of Carbapenem susceptible strain (*E. coli* ATCC 25922) towards a carbapenem disc with a characteristic cloverleaf-like indentation.

Determination of ESBL isolates

The combined-disc tests¹⁷ were performed to evaluate carbapenemase production and to differentiate among *K. Pneumoniae* Carbapenemase (KPC) and Metallobeta-lactamase (MBL) enzymes. We screened the meropenem resistant *K. pneumoniae* isolates for ESBL production by Double Disc Synergy Test (DDST) method¹⁸ according to the criteria of Clinical and Laboratory Standards Institute (CLSI) guidelines¹⁴.

The MIC of the Carbapenem-resistant isolates was determined by broth dilution method according to EUCAST, 2003 guidelines using Meropenem in Mueller-Hinton broth. The potency of meropenem powder (Titan, Biotech Ltd, India) was 86.90. The turbidity of isolates was maintained by comparison with a 0.5 McFarland standard and *E. coli* ATCC 25922 was used as control strain. The

MIC of 16 µg/mL of Meropenem or above were interpreted as resistant^{14,19}.

Genotypic Detection of β-Lactamase specific genes by Multiplex-PCR

DNA was extracted from overnight grown pure cultures of bacterial isolates using Qiagen DNA extraction kit (Qiagen, USA) according to the manufacturer's instructions and the purity was calculated by Spectrophotometer (Titertek, Germany). All isolates were tested for screening of carbapenemase-encoding genes by using PCR technique.

Beta-lactamase genes (*bla*_{CTX-M}, *bla*_{SHV} and *bla*_{TEM}), Ambler class B and D MBLs (*bla*_{VIM}, *bla*_{NDM-1} and *bla*_{OXA}) and serine carbapenemases (*bla*_{KPC}) were detected in the isolates by polymerase chain reaction (PCR). The thermal cycler was set for 36 cycles with an initial denaturation for 5 minutes and final extension for 7 minutes for all PCR. The thermal cycler conditions for specific gene targets were set according to the Table I. PCR products were electrophoresed on 1.2% of agarose gel, stained with Ethidium-Bromide and observed under UV transilluminator (BioRad, USA). Photographs were taken using Gel Doc (BioRad, USA) and the bands were analyzed with 'Quantity One' software (BioRad, USA). The size of the product was calculated observing the relative mobility of the bands in agarose gel compared to the 100 bp DNA ladder (Invitrogen, USA). All sequences of primers are given in the Table I.

Table I: Oligonucleotide Primer sequences used for the detection of antibiotic resistance genes and corresponding PCR conditions

Target genes	Sequence (5'→ 3')	Amplicon size (bp)	Annealing	References
<i>bla</i> _{KPC-1}	TATCGTTGATGTCACTGTATC TTTTCAGAGCCTTACTGCCC	893	51 (1 minute)	(Corbellini et al., 2014)
<i>bla</i> _{KPC-2}	CGTCTAGTTCGTGCTGCTTG CTTGTCATCCTTGTTAGGCG	798	52 (45 sec)	(Nordmann et al., 2011c)
<i>bla</i> _{KPC-3}	ATGTCACGTATCGCCGTCT TTTTCAGAGCCTTACTGCCC	911	55 (45 sec)	(Li et al., 2011)
<i>bla</i> _{VIM-2}	ATTGGTCTATTTGACCGCGTC TGCTACTCAACGACTGAGCG	748	56 (30 sec)	(Lee et al., 2002)
<i>bla</i> _{NDM-1}	GGTTTGGCGATCTGGTTTC CGGAATGGCTCATCAGATC	621	51 (45 sec)	(Nordmann et al., 2011c)
<i>bla</i> _{OXA-64}	ATGGCGATTACTGGATAGATGG AGTCTTGGTCTTGGTTGTGAG	701	52 (30 sec)	(Bali et al., 2010)
<i>bla</i> _{TEM}	TTTCGTGTCGCCCTTATTCC ATCGTTGTGAGAAGTAAGTTGG	403	51 (45 sec)	(Bali et al., 2010)
<i>bla</i> _{CTX-M}	CGCTGTTGTTAGGAAGTGTG GGCTGGGTGAAGTAAGTGAC	569	52 (45 sec)	(Bali et al., 2010)
<i>bla</i> _{SHV}	CGCCTGTGTATTATCTCCCT CGAGTAGTCCACCAGATCCT	293	52 (45 sec)	(Bali et al., 2010)

DNA Sequencing

PCR products were purified using GenElute PCR clean-up kit (Sigma, USA). In cycle sequencing, the ABI-prism BigDye Terminator cycle sequencing ready reaction kit was used for further experiment. The thermal cycle program initiated with initial denaturation at 96°C for 1 minute, followed by 25 cycles of the following steps: denaturation at 96°C for 10 seconds, annealing at 50°C for 30 seconds followed by extension at 60°C for 4 minutes and a single final extension at 60°C for 10 minutes. The purification procedure was as follows: 2µL 3M Na-acetate (pH 4.6) and 50µL of 96.0% ethanol added to each of the tubes containing cycle sequenced product precipitation and centrifuged at 13,000 rpm for 20 minutes. The supernatant was discarded and the precipitate washed using 200µL of 70.0% ethanol and centrifugation at 14,000 rpm for 10 minutes. The tubes were air dried and 16µL of template suppressor reagent (hi-diformamide) was added to each pellet. The content was transferred to a sequencing tube and placed in a thermal cycler at 95°C for 3 minutes for denaturation and analyzed by electrophoresis in an ABI-prism 310 genetic analyzer. DNA was separated through the POP6 contained in a capillary tube and was detected by a laser beam (Manual, ABI Prism USA).

The chromatogram sequencing files were edited using Chromas 2.32 software (Technelysium, Queensland, Australia). The homology of the gene sequences was compared with GenBank database²⁰ using blast²¹. Sequences were edited by bioedit²² and then submitted through Bankitweb-based software.

GenBank accession numbers

Five *bla*_{CTX-M} types 15 were sequenced and submitted to GenBank with the following accession numbers assigned as MH891565 to MH891569. Eleven *bla*_{NDM-1} PCR products were selected randomly for sequencing and were deposited in GenBank with the assigned Accession numbers MH891554 to MH891564 respectively.

Determination of Biofilm formation capacity of isolates

Since the positive correlation of biofilm formation and antibiotic resistance has been demonstrated previously^{12,23}, the biofilm formation capability was analyzed by both qualitative and quantitative assay. The tube adherence method by

Christensen et al.²⁴ were followed. The thin layer of biofilm formed at the inner surface of the test tubes was scrapped with sterile physiological saline impregnated cotton swab and prepared slide was examined under bright field microscope. The biofilm was also measured Quantitatively with a slight modification of the method used by Wakimoto et al.²⁵. A bacterial suspension was prepared and adjusted the optical density of approximately 0.6 at 655nm. Two hundred microliters of suspensions were inoculated into the wells of round-bottom polystyrene microtiter plate (Falcon 3072, New Jersey) and incubated at 28°C for different durations (30 minutes, two hours, eight hours, 24 hours, 48 hours and 72 hours) without shaking. The microtitre plates were washed stained for 30 minutes with 200µl of 0.5% crystal violet solution and 200µl of 95% ethanol was added to each well and the absorbance were read by an ELISA plate reader (BioRad, USA) at 600 nm. A media blank was kept as a negative control (*E. coli* ATCC-25992) and triplicate inoculations were done for each isolate. The result was interpreted according to the formula $BF = AB - CW$ (BF, biofilm formation; AB- stained attached bacteria; CW- stained control wells; All values were at OD 540-600nm²⁶. The value of ≥ 0.300 is considered as strong, range between 0.2 and 0.299 as moderate, range between 0.100-0.199 as weak and of < 0.100 as negative biofilm producer.

Results

This study found 4.33% ($n=28$) out of 647 *K. pneumoniae* isolates from different clinical specimen as carbapenem-resistant by the Modified Hodge test. Most of the isolates were collected from urine (57.14%), followed by blood (17.86%). Almost half of the patients were aged above 60 years. Male patients predominated (60.0%) over females (40.0%). All the isolates were confirmed as *K. pneumoniae* according to cultural characteristics, biochemical screening, and profiling by API 20E (Bio) (Table II). All isolates were resistant to Meropenem with an MIC of at least 64µg/ml up to 512µg/ml (Table III). In addition to carbapenem (Meropenem and Imipenem), most of these isolates showed resistance to Cefepime, Ceftriaxone, Cefixime, Cefuroxime, Amoxicillin, Ampicillin, Nitrofurantoin, Tobramycin, Trimethoprim-Sulfamethoxazole, Amikacin, Gentamycin,

Ciprofloxacin, Chloramphenicol Tetracycline by disc diffusion test (One isolate was sensitive to Amikacin, Gentamycin, Ciprofloxacin, 2 were sensitive to Chloramphenicol and 3 were sensitive to Tetracycline). Fortunately, we found all these isolates sensitive to Colistin. By combined disc test, 82.0% (n=23) of the isolates showed zone enhancement, resembling MBL phenotype. Only two isolates (KPCSQ-18, KPCSQ-25) were phenotypically positive for both MBL-KPC

production and rest of the three isolates were neither MBL nor KPC producer (Table III). For double disc diffusion synergy test, 12 isolates (43.0%) of 28 showed the enhancement of sensitivity zone towards the augmenting disc. By phenotypic tests performed, isolates were categorized into three phenotypic groups, i.e.- ESBL-Carbapenemase, ESBL only and Carbapenemase only.

Table II: Result of the API 20E kit of randomly selected *Klebsiella pneumoniae* isolates

API 20E code	Isolates	Significant taxa identified	% ID
521577317	KPCSQ-9	<i>K. pneumoniae</i>	97.7%
521577317	KPCSQ-16	<i>K. pneumoniae</i>	97.7%
521577317	KPCSQ-18	<i>K. pneumoniae</i>	97.7%

Table III: Phenotypic and molecular characterization of carbapenem resistant *K. pneumoniae* clinical isolates

Strain ID	Clinical diagnosis	Sample	Age (Year)	Quantitative biofilm value	Minimum inhibitory concentration	Modified Hodge test	Double disc synergy test	Combined disc test	Phenotypic resistance pattern	tem	oxa	shv	ctx M	ndm I	Plasmid pattern
KPCSQ1	UTI	U	15-30	0.529	256	+	ESBL	MBL	M+ESBL	+	+	+	-	+	P1
KPCSQ2	UTI	U	60 and above	0.352	256	+	ESBL	MBL	M+ESBL	-	-	-	-	+	
KPCSQ3	UTI	U	45-60	2.07	128	+	Non	MBL	M	+	+	+	+	-	
KPCSQ4	UTI	U	30-45	1.537	128	+	Non	MBL	M	-	+	+	+	-	P5
KPCSQ5	UTI	U	60 and above	2.153	128	+	Non	MBL	M	+	-	+	+	-	P8
KPCSQ6	P	B	0-15	1.39	>256	+	ESBL	MBL	M+ESBL	-	-	-	-	+	
KPCSQ7	WI	SWS	30-45	0.652	>256	+	Non	MBL	M	-	-	+	+	+	
KPCSQ8	UTI	UCST	0-15	1.211	256	+	ESBL	MBL	M+ESBL	+	-	+	+	+	
KPCSQ9	UTI	U	60 and above	1.049	256	+	ESBL	MBL	M+ESBL	+	+	+	-	+	P1
KPCSQ10a	UTI	U	60 and above	1.888	128	+	Non	MBL	M	+	+	+	+	-	P7
KPCSQ10b	UTI	U	60 and above	1.842	128	+	Non	BL	M	+	+	+	+	-	
KPCSQ11	UTI	U	30-45	2.388	128	+	Non	MBL	M	+	+	+	+	-	P5
KPCSQ12	UTI	U	60 and above	1.664	64	+	Non	MBL	M	+	-	+	+	+	P1
KPCSQ13	P	B	60 and above	1.802	256	+	ESBL	MBL	M+ESBL	+	+	+	+	-	P2
KPCSQ14a	UTI	U	45-60	1.967	64	+	Non	MBL	M	+	+	+	+	-	P1
KPCSQ14b	UTI	U	45-60	2.25	64	+	Non	MBL	M	-	-	+	-	-	P3
KPCSQ15a	UTI	U	45-60	2.067	64	+	Non	MBL	M	+	-	+	+	-	
KPCSQ15b	UTI	U	45-60	2.057	64	+	Non	MBL	M	-	-	+	+	+	P1
KPCSQ16	P	SP	60 and above	1.66	128	+	ESBL	MBL	M+ESBL	+	+	+	-	-	P1
KPCSQ17	P	B	0-15	1.291	256	+	ESBL	ampC/ESBL	ESBL	+	+	+	-	+	P6
KPCSQ18	S	B	15-30	1.153	128	+	ESBL	KPC	M+ESBL	-	-	-	-	+	
KPCSQ19	P	SP	30-45	1.656	128	+	ESBL	MBL	M+ESBL	+	-	-	-	+	
KPCSQ20	P	B	60 and above	1.441	128	+	ESBL	ampC/ESBL	ESBL	+	+	+	+	+	
KPCSQ21	ICU	CVLT	60 and above	1.41	256	+	Non	MBL	M	-	-	-	-	+	
KPCSQ22	P	SP	60 and above	2.021	128	+	ESBL	ampC/ESBL	ESBL	-	+	+	-	+	
KPCSQ23	UTI	U	60 and above	2.102	256	+	Non	MBL	M	-	+	+	-	+	P4
KPCSQ24	UTI	U	15-30	1.45	128	+	Non	MBL	M	+	-	+	-	+	P4
KPCSQ25	P	SP	60 and above	0.958	256	+	Non	MBL/KPC	M	-	-	+	-	+	P4

UTI: urinary tract infection, P: pneumonia, WI: wound infection, S: septicemia, ICU: intensive care unit, U: urine, B: blood, SWS: surgical wound swab, UCST: urine catheter suction tip, SP: sputum, CVLT: central venous line tip, M: male, F: female, ESBL: extended spectrum beta lactamase, MBL: metallo beta lactamase, C: carbapenemase, KPC: *Klebsiella pneumoniae* carbapenemase, PP1 through 8: plasmid profile 1 through 8, +: positive and -: negative.

Combination therapy including colistin, carbapenems, etc has been used for the treatment of severe infections caused by CRKP¹ in most cases. Of 25 patients, all received antibiotics before culture and had sensitivity report. Eleven patients received Carbapenem, five patients received cephalosporins and seven patients received both. The rest two patients received Vancomycin and Linezolid respectively. Most of the patients were treated with colistin either alone or in combination with other antibiotics after the culture and antibiotic sensitivity report. Carbapenem continued in case of four patients in combination with either Gentamycin or Vancomycin to complete the continued antibiotic dose or to those cases having lower MIC value. Ceftazidime, Flucloxacillin and Cotrimoxazole were prescribed in three patients respectively.

We performed multiplex PCR analysis targeting the *bla*_{TEM}, *bla*_{SHV} and *bla*_{OXA} genes for genetic determinants of ESBLs and Class D OXA carbapenemase respectively. Of the isolates, 60.0% (n=17) were *bla*_{TEM} positive, 82.0% (n=23) were *bla*_{SHV} positive and 46.0% (n=13) were *bla*_{OXA} positive (Table I). We also targeted *bla*_{CTX-M} and could amplify in 43.0% (n=12) isolates, of which five randomly selected PCR product were sequenced and revealed as *bla*_{CTX-M} type 15. The GenBank accession numbers were assigned as MH891565 to MH891569. More than 60.0% (n=17) isolates were positive for *bla*_{NDM-1}. Among them, 11 PCR products were selected and sequenced (GenBank accession numbers MH891554 to MH891564). Most of the isolates were found to harbor multiple genes. We also performed PCR targeting MBL specific genes. Sixty percent (n=17) of the isolates were found to harbor *bla*_{NDM-1} gene, all the isolates were devoid of *bla*_{VIM-2} or *bla*_{KPC-1} through *bla*_{KPC-4}. Biofilm formation was observed *in vitro* and examined under bright field microscope. Under the microscope clump of bacterial cells were observed and quantified according to formula adapted from Wakimoto et al.²⁵. Twenty-six isolates were found to be strong biofilm former. KPCSQ-1 isolate formed biofilm moderately, but KPCSQ-2 did not form any biofilm at all (Figure 1).

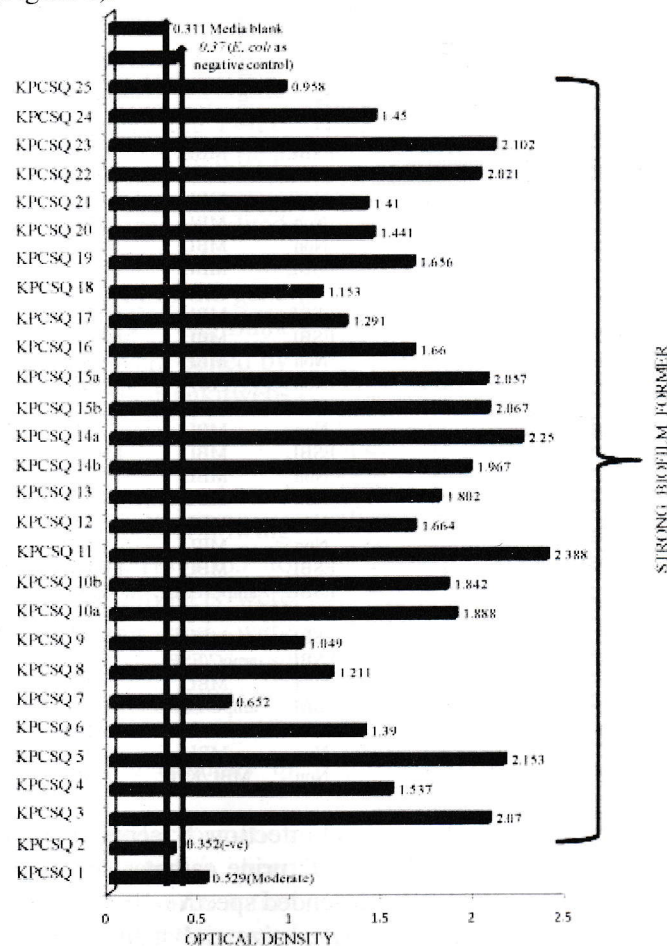


Figure 1: Biofilm forming capability of the isolates

Discussion

Infections due to antibiotic-resistant bacteria are a threat to modern health care and thus the importance of having coordinated and comprehensive national, global actions plans for its control. Carbapenemase-producing bacteria have become a major problem worldwide, which has emerged due to the increased dependence on carbapenems as a last resource to treat bacteria with multidrug resistance. Among the Enterobacteriaceae, CRKP strains have been found as an emerging public health concern because of increasing report of carbapenemase activity and their worldwide dissemination²⁷. Bangladesh is at risk of emerging and re-emerging drug resistant bacterial strains²⁸. In this study, the majority of the CRKP has been isolated from urine (52.0%, n=13), followed by from blood (20.0%, n=5). No correlation was established between the pathotype of *K. pneumoniae* and carbapenemase resistance; however, previous reports also support higher prevalence (around 40.0%) of CRKP in UTI cases²⁹. According to the present study, 12 of the isolates were also positive as ESBL producer, which was similar to a previous report in Bangladesh³⁰. *K. pneumoniae* isolates of the present study were resistant to almost all antibiotics tested. Hence, we call these isolates as extensively drug-resistant (XDR) *K. pneumoniae* according to CDC and ECDC terminology¹. These drugs are most frequently prescribed to treat UTI patients and other infections in Bangladesh²⁸. In this study, 11 out of 28 isolates have an MIC value of $\geq 256 \mu\text{g/ml}$ and 12 isolates have an MIC value of $128 \mu\text{g/ml}$. Previously, in some Bangladeshi uropathogenic isolates, MIC to carbapenems were found to be ranged from $\geq 128 \mu\text{g/ml}$ to $4 \mu\text{g/ml}$ ³¹. Current treatment guidance was based on retrospective observational studies and limited case reports²⁷. However, until now the antibiotics yet to be considered was the last option of drug for treatment of bacterial infection. The emergence of multidrug-resistant (MDR) pathogens seriously threatens to this class of lifesaving drugs⁴. In Bangladesh, three CRKP strains isolated during 2003-2009 were *bla*_{NDM-1} positive. These isolates were found positive for *bla*_{OXA-1} group, *bla*_{CTX-M-1} group (*bla*_{CTX-M-15}) and *bla*_{SHV} genes¹⁰. We found 60.0% (n=17) isolates of this study positive for *bla*_{NDM-1}, of which four isolates were also positive for *bla*_{CTX-M-15}. It was interesting to observe

including these four isolates, *bla*_{CTX-M-15} was found in 43.0% (n=12) of isolates (Table I). *bla*_{OXA-1} found in 46.0% (n=13) of this study's isolates, of which five isolates coharbored *bla*_{NDM-1}, five isolates carried *bla*_{CTX-M-15} and one carried both *bla*_{NDM-1} and *bla*_{CTX-M-15} genes. VIM-1³² or OXA-48 carbapenemases³³ have been found to be co-existed in a widely disseminating clone through Europe³⁴. Moreover, NDM-1 carbapenemase positive isolates were also expressing the CTX-M and OXA-type enzymes³⁵. Among the three major carbapenemases (KPCs, NDMs and OXA-48-like) found in *K. pneumoniae*²⁷, KPCs were absent in our test isolates. Similar studies within Bangladesh and in the neighboring country also confirmed the prevalence of *bla*_{CTX-M-15} among the carbapenemase or ESBL producing isolates in this region^{36,37,38}. Another work in Bangladesh states the *bla*_{NDM-1} gene as the most prevalent (55.0%) gene in carbapenem-resistant Gram-negative uropathogens³¹. Researchers in Dhaka, Bangladesh also detected *bla*_{NDM-1} in 71.0% of nonclinical samples (isolated from water adjacent to hospital region)¹¹, of which *K. pneumoniae* was the predominant bacterial species (44.0%). Coexistence of more than one carbapenemase gene was common in Bangladeshi isolates³¹. Some isolates in our study also showed both the ESBL and carbapenem resistance phenotype. Interestingly, non-ESBL phenotype also carried the genes usually responsible for ESBL phenotype. We found most of the isolates carried different combinations of multiple genes: one isolate (KPCSQ20) carried *bla*_{TEM}, *bla*_{SHV}, *bla*_{OXA}, *bla*_{CTX-M-15}, *bla*_{NDM-1}; four isolate (KPCSQ10a, KPCSQ11, KPCSQ13 and KPCSQ14a) carried *bla*_{TEM}, *bla*_{SHV}, *bla*_{OXA}, *bla*_{CTX-M-15}; three isolates (KPCSQ1, KPCSQ9 and KPCSQ17) carried *bla*_{TEM}, *bla*_{SHV}, *bla*_{OXA}, *bla*_{NDM-1}; one isolate (KPCSQ12) carried *bla*_{TEM}, *bla*_{SHV}, *bla*_{CTX-M-15}, *bla*_{NDM-1}. MBL producing bacteria emerges with virulence factors associated with biofilm production³⁹ in different parts of the world. In this study we also found almost all the isolates to produce strong biofilm. *K. pneumoniae* in a biofilm growth mode resulted in being more resistant to all antibiotics^{12,23}. It might be predicted that these isolates might possess high antibiotic resistance properties *in vivo* condition. Plasmid profiles could demonstrate epidemiological linkage among multidrug-resistant bacteria. Different individuals working in Bangladesh^{38,40}

observed *bla*_{NDM-1} gene in a conjugative plasmid of ~23-24 MDa. The plasmid profile for the test isolates were also analyzed but no correlation was observed with the beta-lactamase profile. Plasmids ranged from 19 MDa to 91 MDa were observed in different isolates, where a 65 MDa plasmid was observed in 10 isolates. No plasmid was observed in 12 isolates indicating chromosome-mediated or inserted genes responsible for the antibiotic resistance. The global emergence of double- and multi-carbapenemase producing hospital-acquired Gram-negative pathogens, are mostly *Klebsiella pneumoniae*, *Pseudomonas aeruginosa* and *Acinetobacter baumannii*.

Conclusion

This finding help to extend the knowledge of the ESBL and multi drug resistance K. pneumonia isolated from clinical samples.

Acknowledgement

We acknowledges the Ministry of Science and Technology, the people's republic of Bangladesh for funding, The Grant number was 160/MEDLS-70.

References

1. Tangden T, Giske CG. Global dissemination of extensively drug-resistant carbapenemase-producing Enterobacteriaceae: clinical perspectives on detection, treatment and infection control. *J Intern Med*. 2015;277:501-12.
2. Bonomo RA, Burd EM, Conly J et al. Carbapenemase - Producing Organisms: A Global Scourge. *Clin Infect Dis*. 2018;66:1290-7.
3. Nordmann P, Cuzon G, Naas T. The real threat of *Klebsiella pneumoniae* carbapenemase - producing bacteria. *Lancet Infect Dis*. 2009; 9:228-36.
4. Nordmann P, Naas T, Poirel L. Global spread of Carbapenemase - producing Entero - bacteriaceae. *Emerg Infect Dis*. 2011; 7:1791-8.
5. Asthana S, Mathur P, Tak V. Detection of Carbapenemase Production in Gram-negative Bacteria. *Journal of Laboratory Physicians*. 2014;6:69-75.
6. NCCLS. Performance Standards for Antimicrobial Susceptibility Testing. Nineteenth informational supplement. In: (CLSI) CaLSI, ed. PA: Wayne; 2009:M100-S19.
7. Lee JJ, Lee JH, Kwon DB et al. Fast and Accurate Large-Scale Detection of beta-Lactamase Genes Conferring Antibiotic Resistance. *Antimicrob Agents Chemother*. 2015;59:5967-75.
8. Yong D, Toleman MA, Giske CG et al. Characterization of a new metallo-beta-lactamase gene, *bla*(NDM-1) and a novel erythromycin esterase gene carried on a unique genetic structure in *Klebsiella pneumoniae* sequence type 14 from India. *Antimicrob Agents Chemother*. 2009;53:5046-54.
9. Castanheira M, Deshpande LM, Mathai D, Bell JM, Jones RN, Mendes RE. Early dissemination of NDM-1- and OXA-181-producing Enterobacteriaceae in Indian hospitals: report from the SENTRY Antimicrobial Surveillance Program, 2006-2007. *Antimicrob Agents Chemother*. 2011; 55:1274-8.
10. Islam MA, Huq M, Nabi A et al. Occurrence and characterization of multidrug-resistant New Delhi metallo-beta-lactamase-1-producing bacteria isolated between 2003 and 2010 in Bangladesh. *J Med Microbiol*. 2013; 62:62-8.
11. Islam MA, Islam M, Hasan R et al. Environmental spread of NDM-1-producing multi-drug resistant bacteria in Dhaka, Bangladesh. *Applied and Environmental Microbiology*. 2017.
12. Vuotto C, Longo F, Balice MP, Donelli G, Varaldo PE. Antibiotic Resistance Related to Biofilm Formation in *Klebsiella pneumoniae*. *Pathogens*. 2014;3:743-58.
13. Shayegani M, Maupin PS, McGlynn DM. Evaluation of the API 20E system for identification of nonfermentative Gram-negative bacteria. *J Clin Microbiol*. 1978;7: 539-45.
14. CLSI CaLSI. Performance Standards for Antimicrobial Susceptibility Testing. 22nd Informational Supplement. 2012:M100-S22.
15. Bauer AW, Kirby WM, Sherris JC, Turck M. Antibiotic susceptibility testing by a standardized single disk method. *Am J Clin Pathol*. 1966;45:493-6.
16. Amjad A, Mirza IA, Abbasi SA, Farwa U, Malik N, Zia F. Modified Hodge test: A

- simple and effective test for detection of carbapenemase production. *Iranian Journal of Microbiology*. 2011;3:189-93.
17. Tsakris A, Poulou A, Pournaras S et al. A simple phenotypic method for the differentiation of metallo-beta-lactamases and class A KPC carbapenemases in *Enterobacteriaceae* clinical isolates. *J Antimicrob Chemother*. 2010;65:1664-71.
 18. Jarlier V, Nicolas MH, Fournier G, Philippon A. Extended broad-spectrum beta-lactamases conferring transferable resistance to newer beta-lactam agents in *Enterobacteriaceae*: hospital prevalence and susceptibility patterns. *Rev Infect Dis*. 1988;10:867-78.
 19. EuCAST ECoAST. Breakpoint tables for interpretation of MICs and zone diameters Version 2. 2012.
 20. Benson DA, Karsch-Mizrachi I, Lipman DJ, Ostell J, Wheeler DL. GenBank. *Nucleic Acids Res* 2005;33:D34-8.
 21. Altschul SF, Madden TL, Schaffer AA et al. Gapped Blast and Psi-Blast: a new generation of protein database search programs. *Nucleic Acids Res*. 1997;25:3389-402.
 22. Hall TA. BioEdit: a user-friendly biological sequence alignment editor and analysis program for Windows 95/98/NT. *Nucleic Acids Symp Ser*. 1999;41:95-8.
 23. Yang D, Zhang Z. Biofilm-forming *Klebsiella pneumoniae* strains have greater likelihood of producing extended - spectrum beta - lactamases. *J Hosp Infect*. 2008;68:369-71.
 24. Christensen GD, Simpson WA, Bisno AL, Beachey EH. Adherence of slime-producing strains of *Staphylococcus epidermidis* to smooth surfaces. *Infect Immun*. 1982;37:318-26.
 25. Wakimoto N, Nishi J, Sheikh J et al. Quantitative biofilm assay using a microtiter plate to screen for enteroaggregative *Escherichia coli*. *Am J Trop Med Hyg*. 2004;71:687-90.
 26. Martinez-Medina M, Naves Pn, Blanco J et al. Biofilm formation as a novel phenotypic feature of adherent-invasive *Escherichia coli* (AIEC). *BMC Microbiology*. 2009;9:202.
 27. Lee CR, Lee JH, Park KS, Kim YB, Jeong BC, Lee SH. Global Dissemination of Carbapenemase - Producing *Klebsiella pneumoniae*: Epidemiology, Genetic Context, Treatment Options and Detection Methods. *Front Microbiol*. 2016;7:895.
 28. Faiz MA, Basher A. Antimicrobial resistance: Bangladesh experience. *Regional Health Forum* 2011;15:1-8.
 29. Nagaraj S, Chandran SP, Shamanna P, Macaden R. Carbapenem resistance among *Escherichia coli* and *Klebsiella pneumoniae* in a tertiary care hospital in south India. *Indian J Med Microbiol*. 2012;30:93-5.
 30. Mehrgan H, Rahbar M. Prevalence of extended-spectrum beta-lactamase-producing *Escherichia coli* in a tertiary care hospital in Tehran, Iran. *Int J Antimicrob Agents*. 2008; 31:147-51.
 31. Begum N, Shamsuzzaman SM. Emergence of carbapenemase-producing urinary isolates at a tertiary care hospital in Dhaka, Bangladesh. *Tzu-Chi Medical Journal*. 2016;28:94-8.
 32. Miro E, Segura C, Navarro F et al. Spread of plasmids containing the bla(VIM-1) and bla(CTX-M) genes and the qnr determinant in *Enterobacter cloacae*, *Klebsiella pneumoniae* and *Klebsiella oxytoca* isolates. *J Antimicrob Chemother*. 2010;65:661-5.
 33. Cuzon G, Ouanich J, Gondret R, Naas T, Nordmann P. Outbreak of OXA-48-Positive Carbapenem-Resistant *Klebsiella pneumoniae* Isolates in France. *Antimicrobial Agents and Chemotherapy*. 2011;55:2420-3.
 34. Potron A, Kalpoe J, Poirel L, Nordmann P. European dissemination of a single OXA-48-producing *Klebsiella pneumoniae* clone. *Clin Microbiol Infect*. 2011;17:E24-6.
 35. Poirel L, Walsh TR, Cuvillier V, Nordmann P. Multiplex PCR for detection of acquired carbapenemase genes. *Diagn Microbiol Infect Dis*. 2011;70:119-23.
 36. Bharadwaj R, Joshi S, Dohe V, Gaikwad V, Kulkarni G, Shouche Y. Prevalence of New Delhi metallo-beta-lactamase (NDM-1)-positive bacteria in a tertiary care centre in Pune, India. *Int J Antimicrob Agents*. 2012; 39:265-6.
 37. Hasan B, Islam K, Ahsan M et al. Fecal carriage of multi-drug resistant and extended spectrum beta-lactamases producing *E. coli* in household pigeons, Bangladesh. *Vet Microbiol*. 2014;168:221-4.
 38. Talukdar PK, Rahman M, Nabi A et al. Antimicrobial resistance, virulence factors and genetic diversity of *Escherichia coli* isolates

- from household water supply in Dhaka, Bangladesh. PLoS One. 2013;8:e61090.
39. Singhai M, Rawat V, Goyal R. Concomitant detection of biofilm and metallo-beta-lactamases production in gram-negative bacilli. Indian J Pathol Microbiol. 2013;56:276-8.
40. Lina TT, Khajanchi BK, Azmi IJ et al. Phenotypic and molecular characterization of extended-spectrum beta-lactamase-producing *Escherichia coli* in Bangladesh. PLoS One. 2014;9:e108735.